Vision-Based Road Rally Crowd Analysis and Agent-Based Simulation

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***Abstract*—Road rallies and other public gatherings alter pedestrian movement on urban streets, causing congestion and restricting road access, yet such changes are difficult to measure consistently through manual observation or ordinary video monitoring. This paper proposes a vision-based framework for analyzing pedestrian movement during road rallies and for exploring alternative crowd conditions through simulation. A pretrained YOLO detector and the ByteTrack tracker are planned to extract pedestrian counts, positions, movement directions, approximate speeds, and density patterns from rally video. These measurements will initialize and calibrate an agent-based simulation in which pedestrian agents follow simple movement rules and avoid one another and obstacles. Three scenarios, namely normal movement, increased crowd density, and a blocked road section, will be compared in terms of pedestrian movement, average simulated speed, congestion, and simulated travel time. This paper presents the design and research background of an undergraduate semester project; no experimental results are reported. The expected contribution is a simple, reproducible workflow that links video-based observation with scenario-based crowd simulation.**

***Index Terms*—Agent-based modeling, ByteTrack, crowd analysis, multi-object tracking, pedestrian detection, YOLO.**

I. INTRODUCTION

Road rallies, protests, and other public gatherings can change the way people use urban streets within a short time. When a large group occupies a road, nearby pedestrians may slow down, change direction, or seek alternative routes, and parts of the road may become inaccessible. Understanding crowd movement in such situations is relevant to public safety, event planning, and traffic management, and has been studied across several disciplines [1].

In practice, however, analyzing crowd movement is difficult. Manual observation depends on the attention of the observer and is hard to repeat consistently, while ordinary video monitoring records events without providing measurements such as pedestrian counts, speeds, or density along a road. Automated methods that convert video into structured movement information are therefore valuable.

Computer vision provides such methods. Pedestrian detection locates people in each frame, although detection becomes harder at low resolution and under partial occlusion [2]. YOLO is a single-network detector that predicts bounding boxes and class probabilities directly from an image and is fast enough for video use [3]. Because detection treats every frame independently, a tracker is required to link detections over time; ByteTrack does this by also matching low-confidence detections, which often belong to partly hidden people, to existing tracks [4].

Observation alone, however, describes only the situation that actually occurred. Agent-based modeling (ABM) represents a system as a set of autonomous agents whose interactions produce collective behavior [5]. In pedestrian simulation, each agent moves toward a goal, avoids other agents, and steers around obstacles; the social force model is a well-known example [6]. Such models allow conditions to be changed and the resulting behavior to be observed.

This paper proposes a system that combines both approaches. Rally video is processed with YOLO and ByteTrack to estimate pedestrian positions, directions, and approximate speeds, which then initialize and calibrate a simplified agent-based simulation of three scenarios: normal movement, increased crowd density, and a blocked road section. The paper describes a planned design rather than an implemented or validated system, and no experimental results are reported. The remainder of the paper is organized as follows: Section II reviews related work, Section III describes the proposed methodology, Section IV discusses expected outcomes and limitations, and Section V concludes.

II. RELATED WORK

A. Pedestrian Detection

Dollár *et al.* evaluated sixteen pretrained detectors on six data sets and found that performance remained weak at low resolutions and for partially occluded pedestrians [2]. YOLO formulates detection as a single regression problem from image pixels to boxes and class probabilities; its authors reported real-time operation at the cost of more localization errors than some competing systems [3]. This trade-off makes a pretrained YOLO model suitable for processing rally video.

B. Multi-Object Tracking

Multi-object tracking estimates bounding boxes and identities of objects across a video [4], which is necessary for measuring speed and direction. SORT combines a Kalman filter with the Hungarian algorithm for fast online association and showed that detection quality strongly affects tracking performance [7]. ByteTrack extends this idea by associating low-score detection boxes with existing tracks instead of discarding them, which helps recover occluded people and reduces fragmented trajectories [4].

C. Crowd Density and Movement Analysis

Zhan *et al.* surveyed crowd analysis from video, covering tasks such as density and motion measurement for crowd management and surveillance [1]. Density can be estimated

directly from images; Zhang *et al.* proposed a multi-column convolutional network that maps an image to a density map and copes with variation in head size caused by perspective [8]. Tracked trajectories can instead provide flow, direction, and speed. In crowded scenes, overlap and occlusion mean that both counts and trajectories remain estimates [1], [2].

D. Agent-Based Crowd Simulation

Even simple agents can produce complex collective patterns [5]. The social force model represents each pedestrian as moving toward a desired direction while being repelled by other pedestrians and obstacles [6]. Helbing *et al.* used a model of this kind to study panic and jamming at exits, showing that crowd behavior can differ sharply from orderly movement [9]. These studies show how local interaction and obstacle-avoidance rules can reproduce congestion that emerges as density rises. Route changes can be represented by assigning agents alternative goals when a path is blocked.

E. Integration of Vision and Simulation

Johansson *et al.* used video recordings and tracking software to determine parameter values of the social force model through evolutionary optimization, and then applied the calibrated model to evacuation, pilgrimage, and urban scenarios [10]. In this way, vision supplies observed positions, directions, and speeds, while simulation can examine situations that were not recorded, such as higher density or a blocked road.

F. Research Gap

Video-based pedestrian analysis and agent-based simulation are both well-established, and earlier work has already linked video data with pedestrian models [10]. This project therefore does not claim to be the first such combination. Its contribution is narrower: a simplified pipeline that uses a pretrained YOLO detector and ByteTrack to obtain movement information from road-rally video for a basic agent-based simulation comparing three road-crowd scenarios. The expected value lies in a clear, reproducible undergraduate-level workflow rather than in state-of-the-art performance.

III. PROPOSED METHODOLOGY

A. System Overview

The proposed workflow, shown in Fig. 1, has two parts. The computer vision stage converts rally video into movement information, and the simulation stage uses that information to initialize and calibrate pedestrian agents whose behavior is compared across three scenarios.

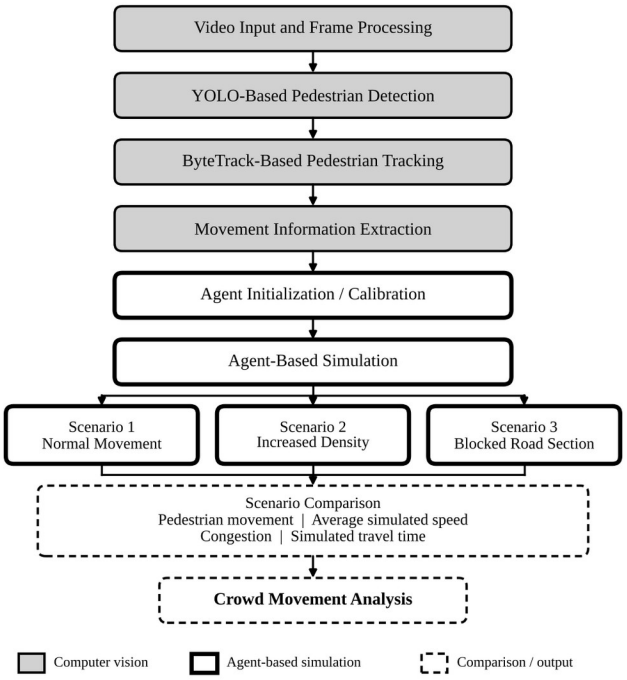


Fig. 1. Proposed workflow of the vision-based road rally crowd analysis and agent-based simulation system (planned; not yet implemented or tested).

B. Video-Based Movement Extraction

Video frames will be processed by a pretrained YOLO detector to locate pedestrians, and ByteTrack will assign persistent identities across frames. From the resulting trajectories, pedestrian counts, positions, movement directions, approximate speeds, and density along the road will be derived. Image displacements will be converted to approximate real-world distances using a known reference length in the scene, and the resulting values are treated as estimates.

C. Agent-Based Simulation

Each pedestrian agent will be initialized with a position, direction, and desired speed taken from the extracted data. Agents will move toward a goal, avoid collisions with other agents, and steer around obstacles using simple social-force-style rules [6]. Parameters will be adjusted so that simulated movement under normal conditions resembles the observed movement, following the calibration idea in [10]. The assumptions of the model will be stated explicitly.

D. Scenarios

Three scenarios are planned, as summarized in Table I. The first reproduces the observed conditions, the second increases the number of agents on the same road, and the third blocks part of the road so that agents must adopt alternative goals.

TABLE I
PLANNED SIMULATION SCENARIOS

Scenario	Simulation condition
Normal movement	Agent count, directions, and speeds as estimated from the video.
Increased density	Larger number of agents on the same road section.
Blocked road	Obstacle on part of the road; agents use alternative goals.

E. Evaluation Metrics

The scenarios will be compared in terms of pedestrian movement patterns, average simulated speed, congestion, and simulated travel time. Congestion will be quantified from local agent density or the proportion of agents moving below a speed threshold, with the exact definition fixed before simulation.

IV. EXPECTED OUTCOMES AND LIMITATIONS

Based on prior work on crowd dynamics [6], [9], increased density is expected to reduce average simulated speed and increase congestion, and a blocked road section is expected to lengthen simulated travel time through rerouting and local crowding. These statements are expectations to be examined, not findings.

Several limitations apply. Detection and tracking accuracy decrease under occlusion and low resolution [2], so extracted counts and speeds remain estimates. The simulation will be a simplified model whose behavior depends on its assumptions and calibration, and the scenarios are hypothetical conditions rather than recorded events. The system has not yet been implemented or validated.

V. CONCLUSION

This paper presented the design and research background of a vision-based framework that links pedestrian detection and tracking in road-rally video with an agent-based simulation of alternative crowd conditions. The proposed workflow is intentionally simple and reproducible, and it builds on established methods rather than claiming a new one. Future work will implement the pipeline, calibrate the simulation on rally video, and report the scenario comparison.

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